

Mission: AI Assisted End-to-End Delivery Automation (Plotcode)

1. Introduction

This document outlines the mission, core modules, and comprehensive workflow for the AI Assisted End-to-End Delivery Automation project, built on the Plotcode platform. The primary objective is to streamline the software development lifecycle from feature request to production deployment, leveraging AI to automate repetitive tasks while maintaining critical human oversight.

2. Project Goal

The overarching goal is to enable project managers and non-technical stakeholders to initiate and oversee development tasks with clear requirements, allowing AI to handle the implementation. This aims to significantly reduce development cycles and empower a broader range of users to contribute to the development process.

3. Core Modules

The AI Assisted End-to-End Delivery Automation system is composed of several interconnected modules, each responsible for a specific stage of the software delivery workflow:

- **Feature Request & Intake:** Handles the submission and initial recording of new feature requests.
- **Human Review & Approval:** Facilitates human validation and approval at critical junctures of the workflow.
- **AI-Assisted Analysis & Planning:** Utilizes AI to analyze repositories, propose implementation plans, and identify risks.
- **Automated Code Generation & Testing:** AI-driven creation of code changes and associated test cases.
- **Automated Validation & CI:** Executes continuous integration processes, including linting, unit tests, integration tests, and security checks.
- **Pull Request Management:** AI-driven creation and management of pull requests.
- **Deployment Management:** Orchestrates deployments to QA/staging and production environments.

- **Monitoring & Closure:** Tracks post-deployment performance and formally closes completed tasks.

4. Delivery Workflow

The following steps detail the end-to-end delivery workflow, emphasizing the interaction between human and AI agents:

1. **Feature Request Submission:** Users submit feature requests via a structured interface (e.g., Slack workflow), providing essential details such as business need, expected behavior, priority, affected service, and requester information.
2. **Request Intake and Tracking:** The submitted request is automatically logged in a tracking system (e.g., ClickUp, Supabase, GitHub issue) with all details and a unique ID.
3. **Initial Human Review:** A product or technical owner reviews the request for validity, feasibility, and alignment with priorities. Only approved requests proceed.
4. **AI-Assisted Repository Analysis:** An AI service analyzes the relevant code repository, identifies impacted files/modules, proposes an implementation plan, and highlights potential risks or dependencies.
5. **Human Approval of Implementation Plan:** A developer or reviewer validates the AI-proposed technical approach before any code changes are initiated.
6. **Branch Creation and Code Changes:** Upon approval, the AI creates a feature branch, applies necessary code changes, and generates or updates relevant test cases.
7. **Automated Validation:** Continuous Integration (CI) processes run linting, unit tests, integration tests, security checks, and build validation. The AI is designed to identify and fix errors autonomously if checks fail.
8. **Pull Request Creation:** The AI generates a pull request (PR) with a summary of changes, test evidence, risk notes, a rollback plan, and a link to the original request ID.
9. **Human Code Review:** Human reviewers assess the PR. The AI may revise based on feedback, but human approval is mandatory for merging.
10. **QA Deployment:** After PR approval and merge, changes are deployed to a QA/staging environment for validation.
11. **Human QA Validation:** The requester or QA owner validates the feature against original requirements. Issues are reported back for fixes.
12. **Production Approval:** Explicit human approval from the responsible owner is required before deployment to production.
13. **Production Deployment and Monitoring:** Approved changes are deployed to production. Logs, metrics, alerts, and user impact are continuously monitored post-

release.

14. **Closure:** Once validated in production, the original request/ticket is updated with deployment status, PR link, release details, and completion notes.

5. Human-in-the-Loop Checkpoints

Critical human intervention points are integrated into the workflow to ensure quality, compliance, and strategic alignment:

- Request approval before implementation.
- Technical plan approval before code changes.
- Pull Request review before merge.
- QA validation before Production deployment.
- Final approval before Production deployment.

6. Important Guardrails

The AI's role is to assist with analysis, code changes, tests, PR creation, and documentation. However, the AI **must not** directly merge code or deploy to Production without explicit human approval. This ensures human accountability and control over critical stages.

7. High-Level Agent Roles

To facilitate this workflow, several specialized AI agents will be employed:

- **Request Intake Agent:** Processes incoming feature requests and logs them.
- **Planning & Analysis Agent:** Performs repository analysis and proposes implementation plans.
- **Code Generation Agent:** Writes and modifies code, including test cases.
- **Validation & Remediation Agent:** Executes automated tests and attempts to fix identified issues.
- **PR Management Agent:** Creates and updates pull requests.
- **Deployment Agent:** Manages deployments to various environments.
- **Monitoring & Reporting Agent:** Tracks production performance and updates task status.

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